

Assignment 1 Data Modeling using ER Diagram and Relational Data Model

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1. Understand the business functions

Figure 1: Create an account

CanadaFlowers.ca 1-888-705-9999 Sign In

Occasions Sentiments Flowers Gifts International RMHC Canada Corporate

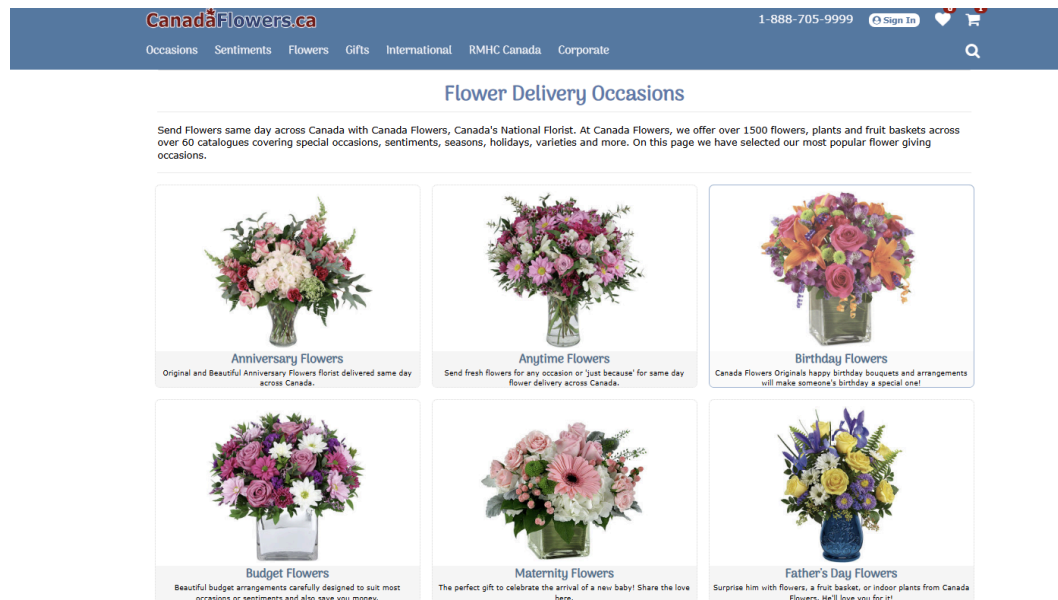
Create Account

Creating an account with Canada Flowers speeds up future ordering. We will save your contact information, along with an address book of recipients (should you wish) for future orders with us. You'll need to use your e-mail address and password next time you log in.

First Name	Street Address
Last Name	City
Email Address	Select a province...
Confirm Email Address	☒ Canada ☐ United States
Phone # Ext	☐ Other
Alt. Phone # Ext	Postal/Zip
Password	Subscribe to Newsletter?
Confirm Password	

- **Purpose:** The Create Account page allows customers to create an account for placing and tracking orders.
- **Key Data Elements:**
 - Customer Identity:
 - First Name
 - Last Name
 - Email Address
 - Account Password.
 - Primary Phone Number
 - Alternate Phone Number.
 - Billing/Home Address:
 - Street Address
 - City
 - Province
 - Country
 - Postal Code.

Figure 2: Searching Flowers Based on Specified Occasions



- **Purpose:** This view serves as the primary navigation and discovery layer for customers. It categorizes the store's inventory into meaningful groups based on the intent of the purchase. One of those categories is Occasion. There are different types of flower arrangements for each occasion such as anniversary, funeral, birthday or even yearly holidays such as fathers/mothers day. The purpose of categorizing flowers by occasion is that it acts as a way to filter out the products according to the user's needs.
- **Key Data Elements:**
 - Occasion Category: (e.g., Birthday, Anniversary, Mother's Day).
 - Relationship to Products: Occasions act as a classification system. A single product can belong to multiple occasions (e.g., a red rose arrangement fits both "Anniversary" and "Valentine's Day"), indicating a many-to-many relationship between products and occasions.

Figure 3: Sentiments

CanadaFlowers.ca

1-888-705-9999

Sign In

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Occasions

Sentiments

Flowers

Gifts

International

RMHC Canada

Corporate

Flower Giving Sentiments

Send Flowers same day across Canada with Canada Flowers, Canada's National Florist. At Canada Flowers, we offer over 1500 flowers, plants and fruit baskets across over 60 catalogues covering special occasions, sentiments, seasons, holidays, varieties and more. On this page we have selected our most popular flower giving sentiments.



Congratulations Flowers

Join the celebration and cheer on their accomplishments today with fresh flowers, bouquets and arrangements.



Friendship Flowers

Friends are the glue that keeps us together in good times and bad. Show a friend today how much they mean to you.



Get Well Flowers

If someone you know needs some cheering up we've got lots of flowers to help!



Love & Romance Flowers

Demonstrate your love and affection by sending flowers from our Love and Romance catalogue.



Sympathy Flowers

We all want to reach out, help and express our sympathies during a time of loss.

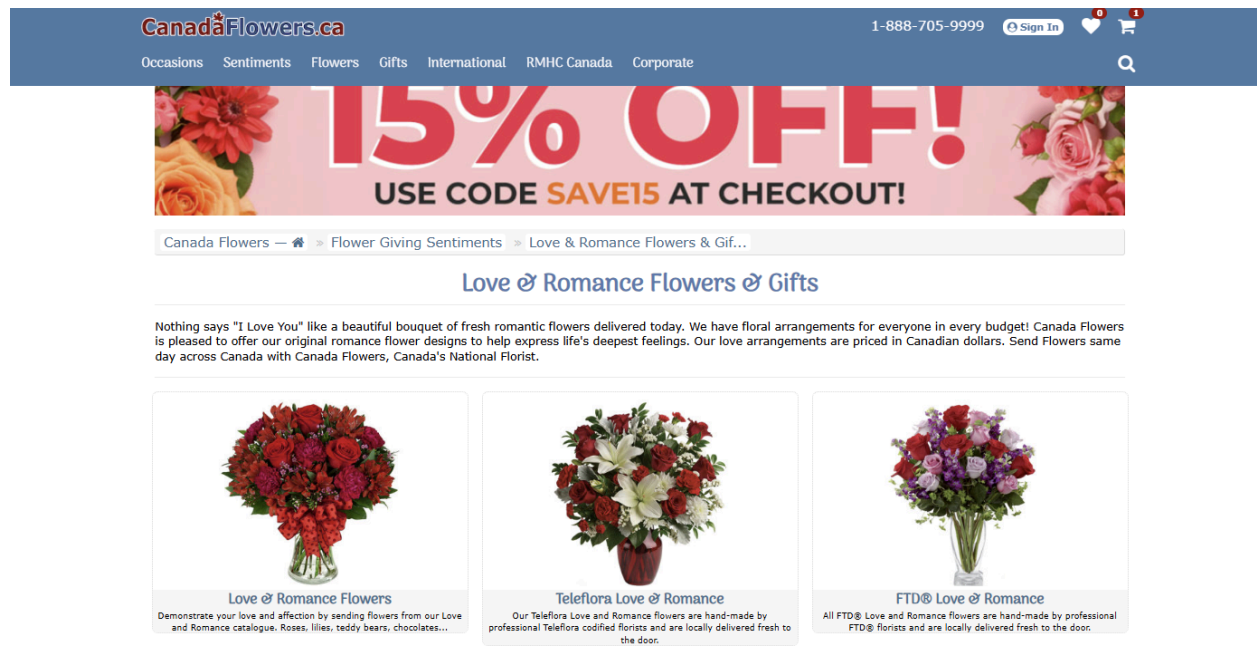


Thank You Flowers

Pay it back by taking a moment to send them some fresh, bright flowers as a great way to simply say Thank you!

- **Purpose:** This is another way to browse products, but based on emotions/feelings (Get Well, Sympathy, Thank You). People can buy flowers for a recovering patient, a romantic partner, a thank you gift, or celebratory sentiments. It is possible that a flower arrangement appears in more than one category. For example, the same product could appear under “Birthday” occasion and “Celebratory” sentiment.
- **Key Data Elements:**
 - Sentiment names
 - Descriptions
 - Category images.
 - Database Link: Also a many-to-many relationship. Products can link to both multiple sentiments and multiple occasions at the same time.

Figure 4: Sub categories



There are also sub-categories inside the two main categories (Occasion and Sentiment). For example, these are the sub-categories under the sentiment category.

- **Purpose:** This page acts as a product catalog grid or listing view. Once a user clicks on a specific sentiment like "Love & Romance", the system filters the database to display only the relevant arrangements matching that theme.

Figure 5: The Product details page

Canada Flowers — » Love & Romance Flowe... » Lasting Love #LV34AA

Lasting Love

♥ Add to Favorites



Demonstrate your deep affection and lasting love with this romantic arrangement of red roses and red miniature carnations in a red glass vase, with added greenery accents.

Our Freshness Guarantee

☐	\$98 \$74 Standard Beautiful flowers. ⓘ alternate photo available
☐	\$121 \$91 Upgraded Recommended. More flowers. Fuller.
☒	\$147 \$111 Premium Our very best version. ⓘ alternate photo available

Add to Shopping Cart

You Could Earn **74** Reward Points ⓘ

What's Included
(Based on 'standard' version, product may vary due to availability.)

 Greenery

 Rose

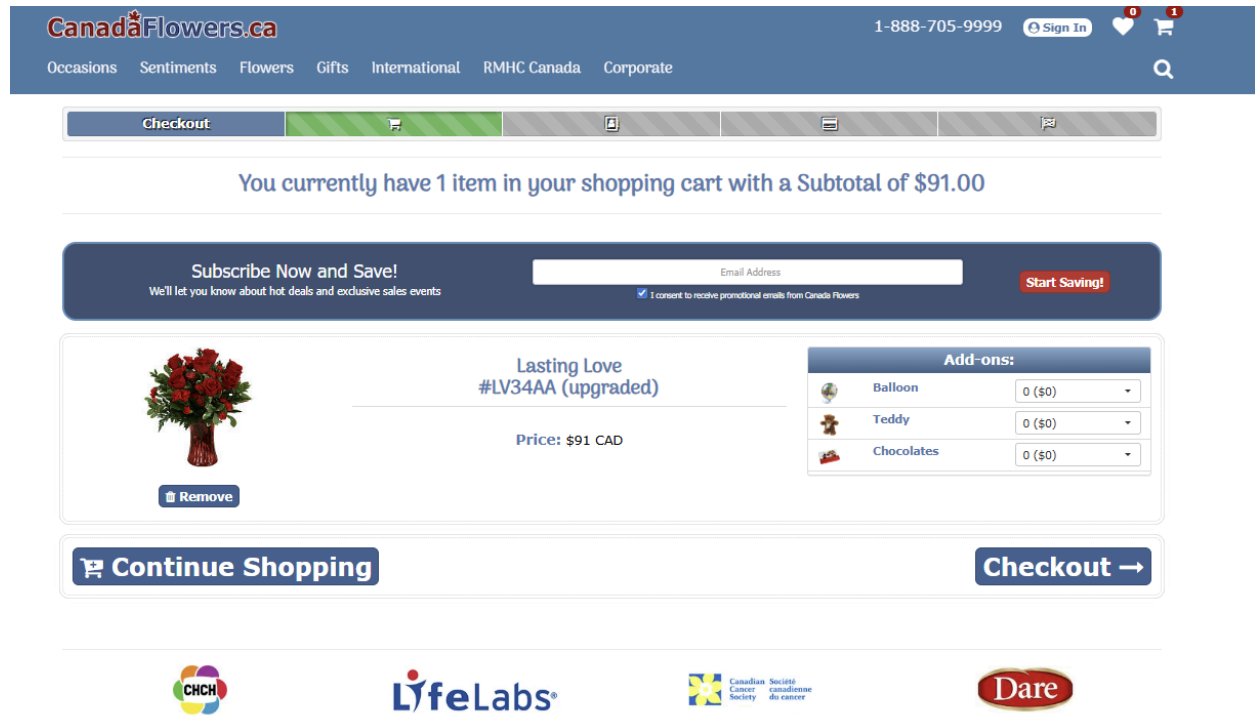
 Mini Carnation

 Container Included

Brought to you by **CanadaFlowers.CA**

- **Purpose:** This view provides comprehensive details about a specific floral item to help customers make an informed purchasing decision and select their preferred configuration.
- **Key Data Elements:**
 - Product Details:
 - Product Name
 - Unique Product ID (e.g., #LV34AA),
 - Text Description
 - Images.
 - Product Versions: Standard, Upgraded, and Premium options.
 - Pricing: Dynamic price points that change based on the selected version (e.g., \$70, \$89, \$100).
 - What's Included: Specific components making up the bouquet (e.g., Rose, Mini Carnation, Greenery, Container).
 - Reward Points: Loyalty points earned upon purchasing the item.

Figure 6: The Shopping Cart



- **Purpose:** This view acts as a temporary review screen before a user begins entering delivery details. It allows the customer to verify their selected item, modify potential add-on choices, see their running subtotal, or jump back to the main catalog.
- **Role of the Shopping cart:** The shopping cart is a temporary holding area. It stores the specific product IDs, selected versions, and chosen add-ons while the user navigates through the checkout steps.
- **Key Data Elements:**
 - Item Count: Shows the total item count currently in the cart and the calculated dynamic subtotal amount (e.g., "1 item... Subtotal of \$91.00").
 - Selected Product Configuration: Shows an image of the selected arrangement, the specific product name and product ID code along with the specific selected upgrade version in brackets (e.g., Lasting Love #LV34AA (upgraded)).
 - Line-Item Price: Displays the explicit price of the chosen product version.
 - Add-ons Panel: A dedicated section displaying available additional purchase entities (Balloon, Teddy, Chocolates).

Figure 7: Delivery Details

The screenshot shows the CanadaFlowers.ca checkout page. At the top, the header includes the website name, phone number (1-888-705-9999), and links for Sign In, a shopping cart with 4 items, and a search icon. Below the header is a navigation bar with links for Occasions, Sentiments, Flowers, Gifts, International, RMHC Canada, and Corporate. The main content area is titled 'Lasting Love' and features a 'Subscribe Now and Save!' banner with an email input field and a 'Start Saving!' button. Below this, a product card displays a bouquet of red roses, the name 'Lasting Love #LV34AA (upgraded)', and the price '\$91 CAD'. To the right of the product card is an 'Add-ons' section with three items: Balloon, Teddy, and Chocolates, each with a '0 (\$0)' price. The 'Delivery Address' section on the left contains fields for 'Select a delivery type...', 'First Name', 'Last Name', 'Street Address', 'Unit # (Optional)', 'City', 'Select a province...', 'Canada', 'Postal/Zip', 'Phone #', and 'Ext.'. Below these fields is a 'Delivery Instructions (Optional)' text area. The 'Delivery Date' section on the right features a calendar for June 2026, with the date 'Thursday, June 4th, 2026' selected. Below the calendar are fields for 'Select an occasion...', 'Card Message' (with a 150-character limit), and 'Card Signature' (with an example: 'Happy Birthday! Love, Robert'). At the bottom of the page are two buttons: '← Cart' and 'Continue to Payment →'.

CanadaFlowers.ca 1-888-705-9999 Sign In 4

Occasions Sentiments Flowers Gifts International RMHC Canada Corporate

Checkout

Lasting Love

Subscribe Now and Save!
We'll let you know about hot deals and exclusive sales events.

Email Address Start Saving!

Lasting Love
#LV34AA (upgraded)
Price: \$91 CAD

Remove

Add-ons:

- Balloon 0 (\$0)
- Teddy 0 (\$0)
- Chocolates 0 (\$0)

Delivery Address

Select a delivery type...

First Name Last Name

Street Address

Unit # (Optional)

City

Select a province...

Canada

Postal/Zip

(Phone number must be for the receiver of the order)

Phone # Ext.

Delivery Instructions (Optional)

You have 150 characters left.

Delivery Date

June 2026

Su	Mo	Tu	We	Th	Fr	Sa
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Your delivery date is:
Thursday, June 4th, 2026

Select an occasion ...

Card Message

You have 150 characters left.

Card Signature

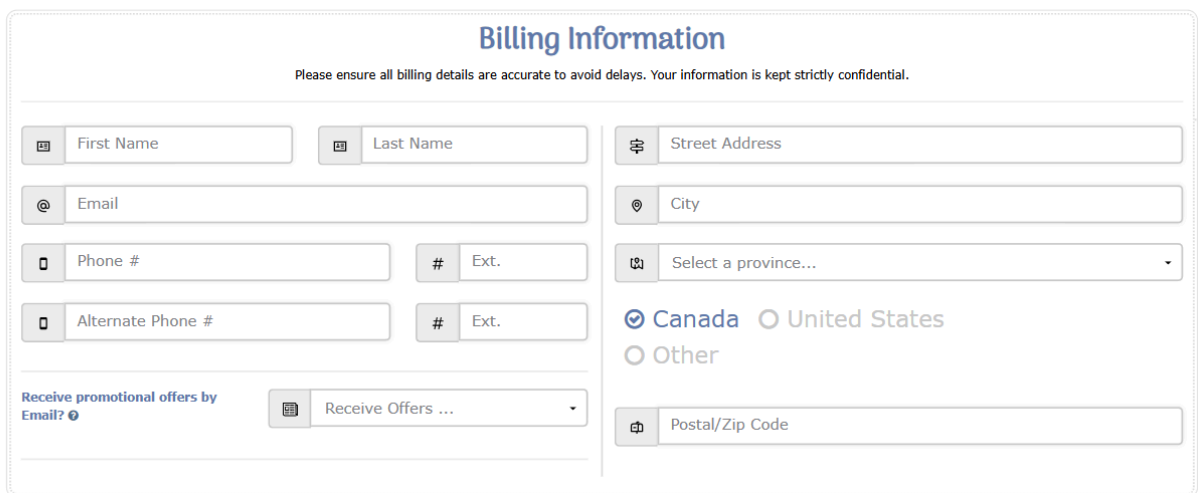
e.g. Happy Birthday! Love, Robert

← Cart Continue to Payment →

- **Purpose:** This is the screen right after clicking checkout from the main shopping cart. It serves as the data collection interface for logistics and delivery. It allows the user to review what is in their cart at the top while filling out the recipient's information below it.
- Key Data Elements:
 - Delivery Address Fields:
 - Delivery Type dropdown (e.g., Residence, Business, School, Hospital)
 - First Name
 - Last Name
 - Street Address
 - Unit Number
 - City
 - Province
 - Country (defaulted to Canada)
 - Postal/Zip code

- Phone Number
- Delivery Instructions: A text area field meant for delivery specifics.
Delivery Date Picker: An interactive calendar where a required delivery date is picked.
- Card Configuration Fields:
 - Specific occasion for the gift card.
 - Card Message
 - Card Signature

Figure 8: Billing Information



The form is titled "Billing Information" in blue. Below the title is a disclaimer: "Please ensure all billing details are accurate to avoid delays. Your information is kept strictly confidential." The form is divided into two main sections. The left section contains fields for "First Name", "Last Name", "Email", "Phone #" (with a "# Ext." label), and "Alternate Phone #" (with a "# Ext." label). Below these is a checkbox for "Receive promotional offers by Email?" and a dropdown menu for "Receive Offers ...". The right section contains fields for "Street Address", "City", "Select a province..." (a dropdown menu), and "Postal/Zip Code". Below the province dropdown are three radio buttons for "Canada", "United States", and "Other".

- **Purpose:** This view captures the personal and address details of the purchaser (the sender) during the checkout sequence. It keeps the billing party's information decoupled from the delivery destination data, ensuring proper transaction tracking, invoicing, and card processing. In our data model, this explicitly defines the attributes for a Sender or Billing Profile. It establishes that an order relies on two distinct address tracking mechanisms: one for where the bill goes (this view) and one for where the flowers go (the recipient view).
- **Key Data Elements:**
 - Sender Identity:
 - First Name
 - Last Name
 - Email address.
 - Primary Phone number
 - Alternate Phone number.
 - Billing Address Details:
 - Street Address
 - City
 - Province
 - Country(Canada, United States, and Other),
 - Postal/Zip

Figure 9: Payment Method and Order Summary

Payment Method



Select payment type...

Credit Card Number

Expiry Month

Expiry Year

Security Digits

Name on Card

Order Summary

Price	\$91.00
Delivery	\$15.00
Subtotal	\$106.00
GST/HST	\$13.78
Total Price	\$119.78

*All prices are in Canadian dollars

You Could Earn 91 Reward Points

Enter Promo Code

Have Gift Card Code?

← Delivery Info

Review Order

Place Order

- **Purpose:** This view acts as the final confirmation and checkout terminal. It allows the customer to securely enter financial credentials, review an exact mathematical breakdown of costs, apply promotional text codes, and formally execute the order transaction.
- **Key Data Elements:**
 - Payment Method:
 - Payment Type (e.g., Visa, Mastercard, Amex, Discover, PayPal)
 - Credit Card Number
 - Expiry Month
 - Expiry Year
 - Security Digits
 - Name on Card.

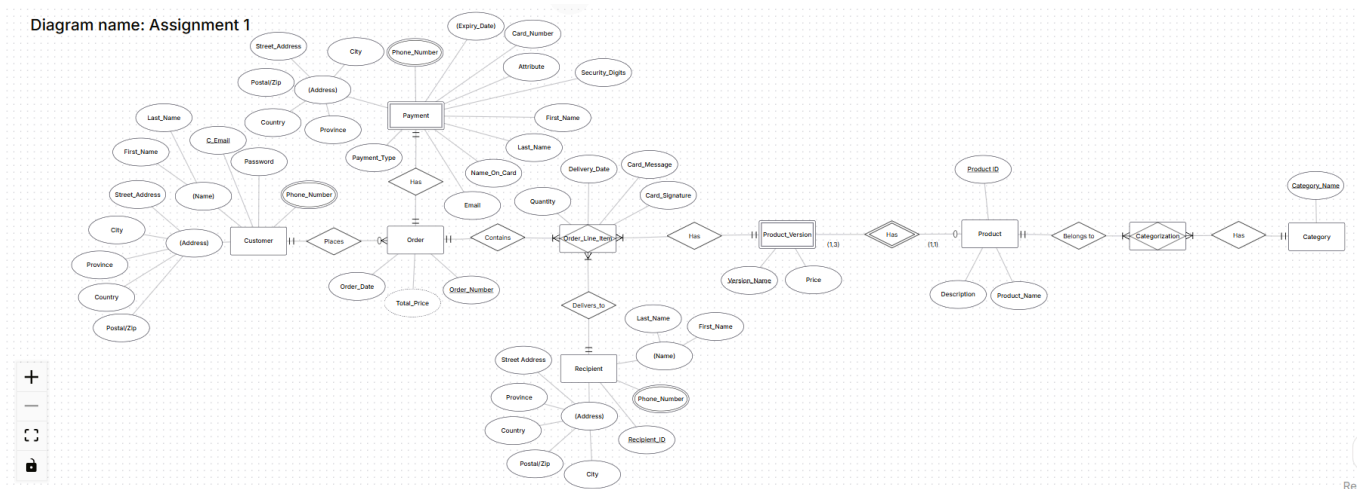
2. State your business rules and assumptions

To build the E-R model, I have established the following rules based on the CanadaFlowers.ca site structure.

- Customers and Accounts:
 - A customer can place an order as a guest or by creating a permanent account.
 - Each registered customer is identified by a unique email address.
 - Registered customers can save a "book of recipients" (an address book) to reuse delivery info for future orders.
- Products and Categorization:
 - Flowers are organized into two main types of categories: Occasions (like Birthdays) and Sentiments (like Love & Romance).
 - A single product can belong to multiple categories (e.g., one bouquet can be for both a "Birthday" and "Love & Romance"), and each category contains many products. This is a Many-to-Many relationship.
- Product Versions and Pricing:
 - Every bouquet has a unique Product Code (e.g., #LV34AA).
 - Each product comes in three specific versions: Standard, Upgraded, and Premium.
 - The price is not fixed to the product itself but is determined by which Version the customer selects.
 - Products can have optional Add-ons (Balloons, Teddies, Chocolates) that each have their own quantity and price.
- Orders and Fulfillment
 - An Order is created once a payment is processed.
 - Each order is linked to one Sender (the person paying).
 - An order can contain multiple products, and each product in that order is sent to a specific Recipient.
 - Even if the Sender and Recipient are the same person, the database must store their information separately (Billing Address vs. Delivery Address).
 - For every item sent, the system must capture a specific Delivery Date, an Occasion for the card, and a custom Card Message.
- Payments and Totals:
 - An order must track the financial breakdown: Price, Delivery Fee, Subtotal, Taxes (GST/HST), and the final Total Price.
 - Each order calculates Reward Points based on the items purchased, which are credited to the customer's account.

3. Build an E-R model

Diagram name: Assignment 1



1. Core Entities

- **Customer**: Represents the registered user. Identified by C_Email. It includes composite attributes for Name and Address, and a multivalued Phone_Number to match real-world registration.
- **Order**: The central transaction record. It stores the Order_Number (Primary Key), the total price, and the date the purchase was finalized.
- **Payment**: A weak entity tied to the Order. It captures the specific financial authorization details (Card Number, Expiry, Security Digits) and the Billing Address provided at the final checkout step.
- **Recipient**: Represents the delivery target. Each recipient is uniquely identified by a Recipient_ID and includes a full shipping address and contact phone number.
- **Product**: The master catalog entry for an arrangement (e.g., "Red Rose Bouquet"). Identified by Product_ID.
- **Product_Version**: A weak entity representing the "Standard," "Upgraded," or "Premium" tiers of a product. It depends on the Product for its identity.
- **Category**: Represents the categories (Occasions and Sentiments).

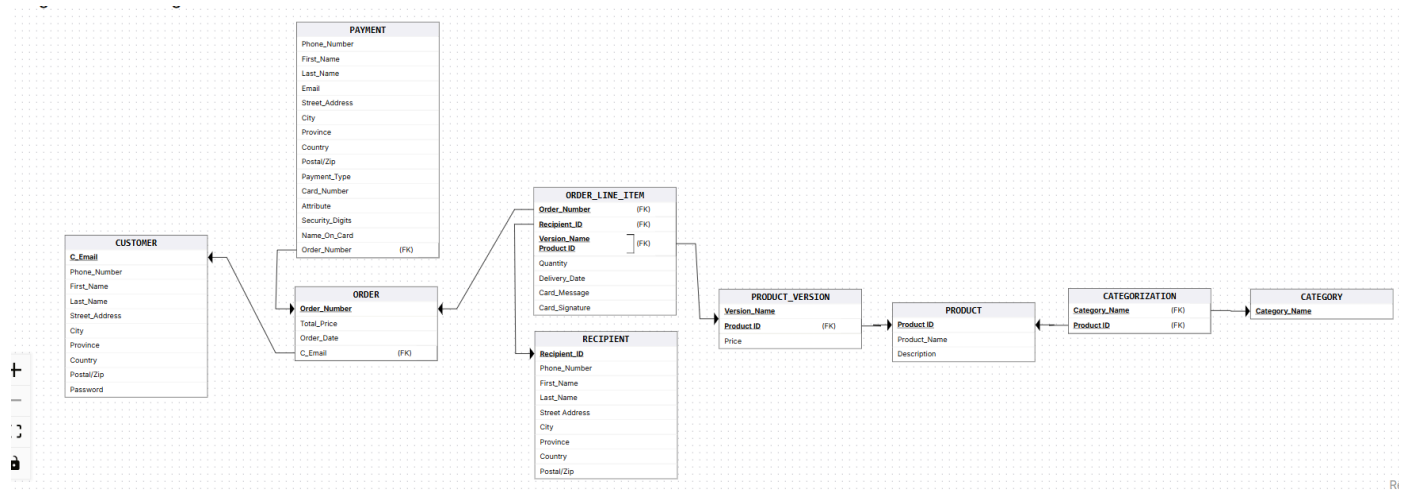
2. Associative Entities

- **Order_Line_Item:** The most critical component. It bridges Order and Product_Version. Because each item in the cart can be personalized, it stores the Quantity, Delivery_Date, Card_Message, and Card_Signature specifically for that one item.
- **Categorization:** A bridge that allows Products to be listed under multiple Categories (e.g., a bouquet can be both "Birthday" and "Romantic").

3. Key Relationships & Cardinality

- **Customer to Order:** A 1:N relationship. One customer can place many orders, but each order belongs to only one customer.
- **Order to Order_Line_Item:** A 1:N identifying relationship. One order can contain many items.
- **Order_Line_Item to Recipient:** A N:1 relationship. Each specific line item is sent to exactly one Recipient, though one Recipient can receive many different items.
- **Product to Product_Version:** A 1:N identifying relationship where the parent product provides various size/price tiers.

4. Convert your E-R model to relational tables



5. Normalization check

1. Functional Dependencies and Assumptions: To assess the Normal Form of the database, the following functional dependencies (FDs) are identified based on the business rules of the CanadaFlowers storefront:

- Customer Identification: C_Email uniquely determines all personal and account attributes (First_Name, Last_Name, Street_Address, Password).
- Order Integrity: Order_Number uniquely determines the transaction metadata (Total_Price, Order_Date) and the customer who placed it (C_Email).
- Product Catalog: Product_ID uniquely determines the Product_Name and Description.
- Item Personalization: The composite key (Order_Number, Product_ID, Version_Name) uniquely determines the Quantity, Card_Message, and the specific Recipient_ID for that item.

2. Normal Form Analysis: Every table in the proposed relational schema has been analyzed to ensure data integrity and the removal of redundancies.

- 1st Normal Form (1NF): All tables are in 1NF as all attributes are atomic (e.g., Names and Addresses are decomposed into individual fields) and there are no repeating groups or multi-valued attributes within a single record.

- 2nd Normal Form (2NF): All tables are in 2NF. In tables with composite primary keys, such as ORDER_LINE_ITEM and PRODUCT_VERSION, every non-key attribute (like Price or Quantity) is functionally dependent on the entire composite key, rather than a subset of it.
- 3rd Normal Form (3NF): All tables are in 3NF because there are no transitive dependencies. Every non-key attribute depends strictly on the primary key. For example, in the ORDER table, while the C_Email depends on the Order_Number, no additional customer details are stored there, ensuring that non-key attributes do not depend on other non-key attributes.

3. Conclusion

The database successfully achieves Third Normal Form (3NF). By separating multivalued attributes (like Phone Numbers) into their own tables and utilizing associative entities with composite keys to manage many-to-many relationships, the design eliminates data redundancy and prevents update, insertion, and deletion anomalies. This structure ensures that the data remains consistent and scalable as the number of orders and products grows.